DATA STRUCTURES

LAB PROGRAMS

22.Dijkstar’s algorithm

```
#include <stdio.h>
#define V 4
#define INF 9999
int main() {
    int graph[V][V] = {
        {0,4,0,0},
        {4,0,8,2},
        {0,8,0,5},
        {0,2,5,0}
    };
    int dist[V], visited[V]={0};
    for(int i=0;i<V;i++)
        dist[i]=INF;
    dist[0]=0;
    for(int count=0; count<V-1; count++) {
        int min=INF,u;
        for(int i=0;i<V;i++)
            if(!visited[i] && dist[i]<min) {
                min=dist[i];
                u=i;
```

```
        }
      visited[u]=1;
      for(int v=0;v<V;v++) {
         if(!visited[v] && graph[u][v] &&
            dist[u]+graph[u][v] < dist[v])
             dist[v]=dist[u]+graph[u][v];
      }
   }
   printf("Vertex Distance\n");
   for(int i=0;i<V;i++)
      printf("%d\t%d\n",i,dist[i]);
   return 0;
}
```

Output

```
Vertex Distance
0	0
1	4
2	11
3	6


=== Code Execution Successful ===
```